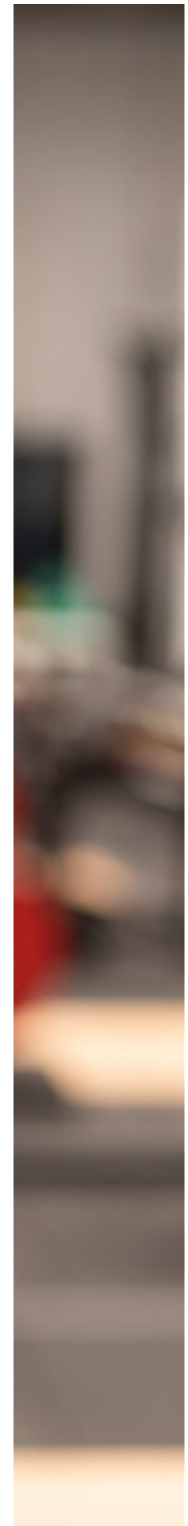

June 17, 2024

Rethinking VBT for real-world coaching

Velocity based training devices are no longer essential for getting great bar speed data. It's time for velocity based training to grow and evolve.

Thomas McInnis
Co-founder

> *"VBT is so important for our lifting at our training centre, but it's just impossible to haul a full kit around with us when we are competing*

> *internationally. It makes it tough to integrate it fully in my programming for the team."*

We recently heard this complaint (third hand) from a highly respected S&C coach of a national level team. It completely aligns with our own experience using VBT as coaches: VBT is amazing, **but implementing VBT is much harder than it should be**.

That is why we are developing Metric, a mobile app that measures velocity without any additional equipment.

We operate a select entry gym for athletes in Melbourne, Australia. In the fast-paced private gym environment we have to balance our use of innovative sports science technology with practical "getting things done" coaching.

Many (many!) years ago we started using a Linear Position Transducer (LPT) device to start working with VBT. We loved what VBT data was telling us about our athletes' performance, but very quickly it became too hard to use, and often got in the way of good coaching.

Building a better VBT device

We tried several LPT systems and accelerometer devices hoping to find something that would work for us. But none of them gave us the right data at the right times, and they all disrupted our coaching and athlete engagement in some way.

We had a list of complaints around these traditional VBT devices: connecting Bluetooth was often unreliable, setting up a workout station or transitioning between the squat rack and bench took way too long, and the software was confusing.

Being a technically minded team of DIY enthusiasts, we started a little side-project to build a VBT device of our own.

First we devised our own rudimentary LPT that we prototyped with springs from a tape measure for the mechanism! Ultimately we found that attaching a physical string to the bar and securing the measurement unit to the floor was just too cumbersome in a busy gym.

Next we actually attempted to use sonar. We had used sonar before in some other experiments successfully. It can be very reliable at establishing distance data, and requires very little power to run making it great for building small battery powered devices.

Our design was actually pretty cool; it was a pendulum box magnetically attached to the barbell pointed downwards, measuring the change in distance to the ground.

Quickly it became obvious that there were problems getting the thing to maintain its angle pointing *exactly* downward without swaying. (The *FLEX* device from GymAware overcomes this by pointing their infrared beams 360º around the barbell... smart!)

Hardware is the problem

Of course, building a new hardware device wasn't really solving our actual complaints about VBT.

If we built a better device, it would still have to be attached and detached from barbells, sync with a paired iPad or iPhone, be kept charged, require firmware patches, and just generally get in the way.

Really, the problem wasn't something we could fix by improving the hardware.

As long as we had to deal with extra hardware, it was going to get in the way of good coaching in our busy gym. *(We also believe that hardware is the reason more individual lifters aren't using VBT in their own training, despite how amazing it is.)*

We needed a solution that eliminated the device completely.

This realisation reset our thinking, and so we... shelved the project! There were many other tech challenges we were trying to solve for the gym, so our attention moved to those.

It's the software, obviously

But David (our incredible software engineer) couldn't stop thinking about the VBT problem. About two years ago he suggested something while we were training in the gym that blew our minds:

What if you could record video of a lift on your phone, and automatically measure velocity data from the video?

Of course we had to ask: is something like that technically possible?

Could it be accurate? Would the user have to pre-measure their setup and range of motion? Would they need to have markers or some additional object attached to the barbell?

David was sure it could be done, and convinced that it could be achieved without any additional markers or measurement.

But more important than our practical concerns, we began to imagine what could be done if it was possible:

- You could have VBT built into a workout tracking app; no more switching between apps or transferring data
- You could integrate VBT data into tools for analysing lifting form and consistency
- Coaches could have access to VBT monitoring for their team while on the road
- We could instantly fit out our entire gym with velocity tracking capabilities by simply downloading an app
- Athletes training away from their home gym could send video and velocity data directly to their coach for feedback

Basically, what if we could put VBT in your pocket?

Velocity Based Training in your pocket

After 18 months of research and development we have built what we believe will become the new standard of measuring velocity.

Metric is a computer vision tool that analyses video and determines the velocity of your lifts. It works with video recorded from a fixed position, taken roughly side-on to a barbell lift using standard size weight plates.

No convoluted set up, no additional equipment to carry around. Just position your phone side-on to the barbell, hit record, and lift. You can see the prototype in action below!

VBT Coach - Velocity Based Training Made Practical

Watch on

This unique tech will be coming to the iPhone in the **Metric VBT app** soon. We think this is a huge deal for VBT practitioners, as well as lifters and coaches who have never had access to VBT before.

Our initial app design is pretty conservative: Metric will allow you to measure a single set of a barbell movement and get reliable data for Velocity (peak and mean), Range of Motion and fatigue. We intend to quickly push out updates for power metrics, bar path analysis, and much more.

Ultimately we want to build something that empowers athletes to perform better, and coaches to deliver great coaching.

That means a tool that can stay out of your way, helping you optimise your training or coaching in real time with context-rich data and actionable intel.

As coaches and lifters ourselves we are incredibly excited by what

Metric can unlock for us. If you are exited too, make sure to join our mailing list. We will be sharing insights and updates as we make this vision for VBT a reality!

Update: Metric is now available on iOS for free.

Get it at this link →

← Return to blog home

By lifters for lifters

Metric was built with lifters and coaches front of mind.

Focused analytics and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.

METRIC
Barbell velocity tracking

Metric App

Metric for Coaches

Metric User Guide

Validation

Available data & metrics

Supported devices

Blog

Release Notes

Support

Privacy policy

Terms & Conditions